Exp No:3) Experiment to understand the data preprocessing in Data science

Code:

```
import numpy as np
import pandas as pd

df = pd.DataFrame({
    'Country': ['India', 'France', 'India', 'France', 'India'],
    'Age': [29, np.nan, 24, 35, 30],
    'Salary': [32000, 40000, 37000, np.nan, 51000],
    'Purchased': ['Yes', 'No', 'Yes', 'No', 'Yes']
})

print("Original Dataset:")
print(df)

print("\n\nDataFrame Information:")
print("\n\nDataFrame Info:")
print(df.info())

print("\n\nCountry Mode:")
print(df['Country'].mode()[0])

print("\n\nFirst Mode Value:")
```

```
print(df['Country'].mode()[0])

df['Country'] = df['Country'].fillna(df['Country'].mode()[0])
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())

print("\n\nDataset After Filling Missing Values:")
print(df)

print("\n\nOne Hot Encoding:")
print(pd.get_dummies(df['Country']))

updated_dataset = pd.concat(
   [pd.get_dummies(df['Country']), df.iloc[:, 1:]],
   axis=1
)

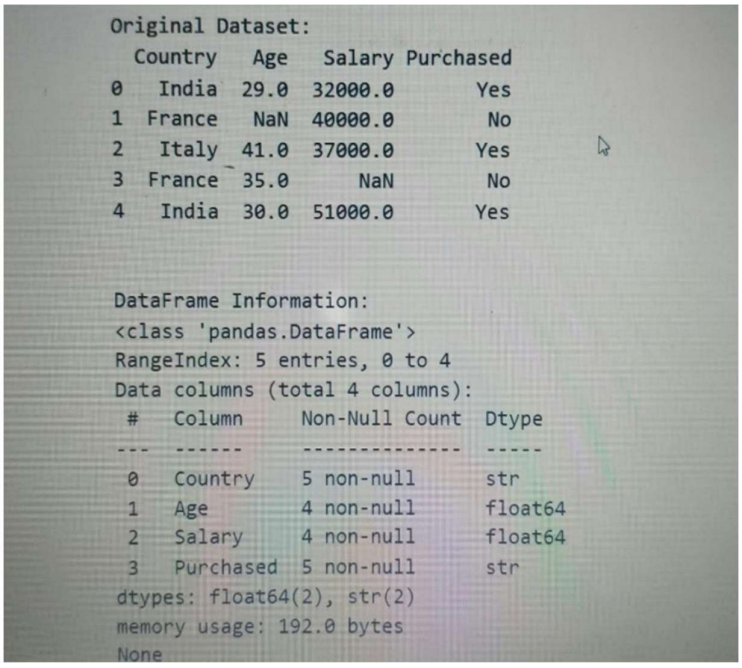
print("\n\nUpdated Dataset:")
print(updated_dataset)

print("\n\nDataFrame Information After Preprocessing:")
print(df.info())
```

```
updated_dataset['Purchased'] =
updated_dataset['Purchased'].replace(
    ['NO', 'Yes'], [0, 1]
)
```

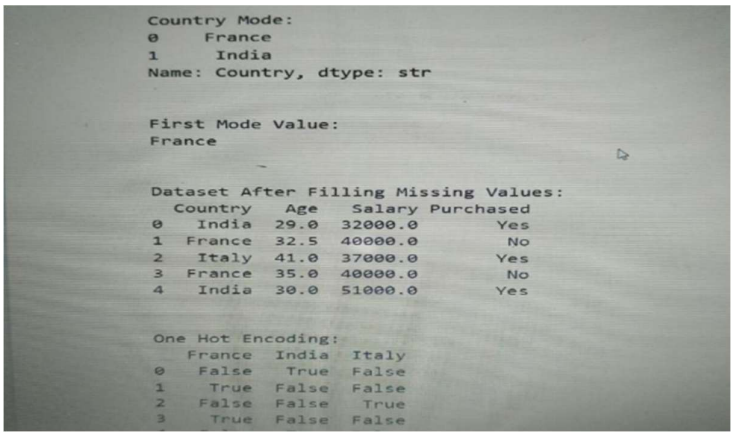
```
print("\n\nFinal Dataset:")
print(updated_dataset)
```

Output:



```
Original Dataset:
  Country   Age   Salary Purchased
0   India  29.0  32000.0       Yes
1  France   NaN  40000.0        No
2   Italy  41.0  37000.0       Yes
3  France  35.0      NaN        No
4   India  30.0  51000.0       Yes

DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    5 non-null      str
 1   Age        4 non-null      float64
 2   Salary     4 non-null      float64
 3   Purchased  5 non-null      str
dtypes: float64(2), str(2)
memory usage: 192.0 bytes
None
```

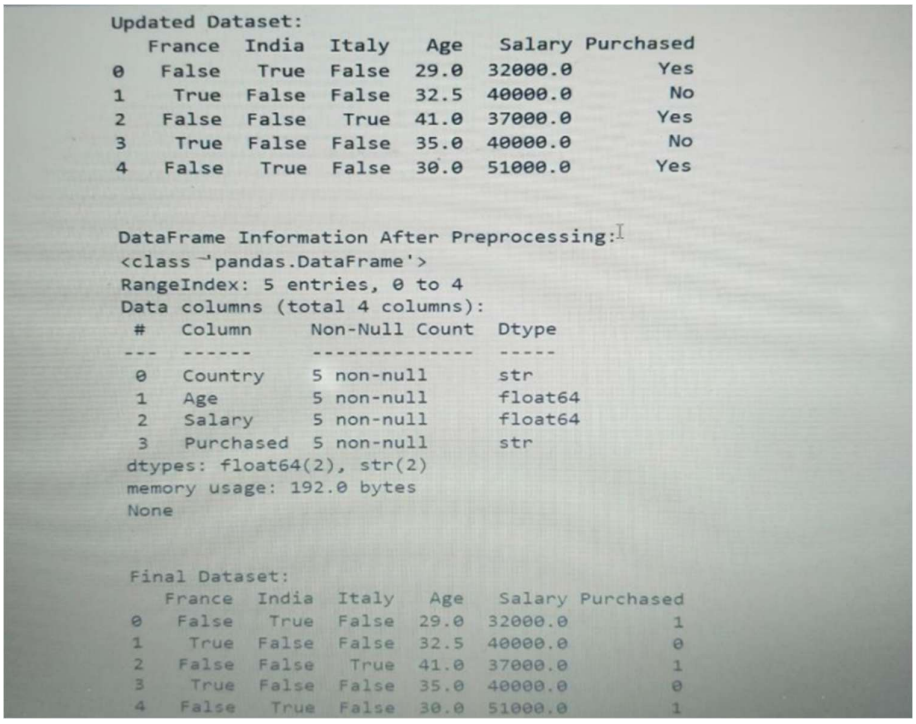


```
Country Mode:
0    France
1     India
Name: Country, dtype: str

First Mode Value:
France

Dataset After Filling Missing Values:
  Country   Age   Salary Purchased
0   India  29.0  32000.0       Yes
1  France  32.5  40000.0        No
2   Italy  41.0  37000.0       Yes
3  France  35.0  40000.0        No
4   India  30.0  51000.0       Yes

One Hot Encoding:
   France  India  Italy
0   False   True  False
1    True  False  False
2   False  False   True
3    True  False  False
```



```
Updated Dataset:
   France  India  Italy   Age   Salary Purchased
0   False   True  False  29.0  32000.0       Yes
1    True  False  False  32.5  40000.0        No
2   False  False   True  41.0  37000.0       Yes
3    True  False  False  35.0  40000.0        No
4   False   True  False  30.0  51000.0       Yes


DataFrame Information After Preprocessing:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    5 non-null      str
 1   Age        5 non-null      float64
 2   Salary     5 non-null      float64
 3   Purchased  5 non-null      str
dtypes: float64(2), str(2)
memory usage: 192.0 bytes
None


Final Dataset:
   France  India  Italy   Age   Salary Purchased
0   False   True  False  29.0  32000.0         1
1    True  False  False  32.5  40000.0         0
2   False  False   True  41.0  37000.0         1
3    True  False  False  35.0  40000.0         0
4   False   True  False  30.0  51000.0         1
```

Result:

Hence, the data pre-processing in data science is analysed.